What a Small Role-Prompted Aggregation Pipeline Does with Instructed Phrases, Planted Figures and Caveats: A Registered Component Study on One 20B Model

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Draft v0.5, October 2026. Adds the equal-cost accuracy experiment (68); supersedes v0.4.

Abstract

Mixture-of-Agents pipelines send a task to several proposer models and have an aggregator write the final answer. We study a much smaller relative of that design on local hardware: one 20B model writes the answer from three contributions that, in most experiments, come from the same model under one-sentence role prompts. Across 68 numbered experiments registered before they ran, the last seven after an internal review, five groups of results hold. (1) An instruction to label estimates with a named phrase produces the phrase in two thirds of answers (378 runs), and a synonym chosen the day before behaves the same. (2) Two local LLM judges prefer the pipeline's answer to a single answer from the same model (all-pairs scores 0.61 and 0.58) and mark down the instructed phrase; an attempt to add judges from other model families failed, because three of four chose the first-listed answer in at least 98% of judgments. (3) On 61 questions with computed answers, the aggregator's answer follows a figure stated in the question over a contribution that misquotes it (0 of 122 runs against 112 of 122 when the contribution is the only source) and does not mention the misquote; when two contributions give different figures and the question gives none, the answer follows the one listed first about three times in four. (4) Six reader models from five developers change their decision when a decisive sentence is appended to a report ($+0.52$ pooled over four new readers, 11 items), without being told to read attachments, and with a loss of about 0.12 when it is one line among ten. (5) On 136 public ten-option questions (MMLU-Pro law, health and business) screened so that one answer from the model is right about half the time, the pipeline was right in 0.566 of runs against 0.496 for one answer ($+0.070$ $[+0.018,\ +0.123]$), but a vote of four plain answers reached 0.561 at the same number of calls, and the same aggregator given three plain drafts 0.544: the role prompts are not shown to add accuracy ($+0.022$ $[-0.034,\ +0.078]$). A design assembled from these observations showed no detectable difference in judge preference from the plain pipeline at about three and a half times the model calls, was 0.110 $[0.044,\ 0.176]$ less accurate than the plain pipeline on the public questions, and in its live chain, scored a second time after a tool fault was repaired, a second source raised right answers from 2 to 19 of 61 and the conflict appeared in 2 of 61 final answers. Two questions stayed unanswered: the sentence-level judge used throughout agrees little beyond chance and could not be compared with a person, and a test of whether caveats are conveyed less often than other sentences was blocked when its judges failed one of four checks. No person labelled, judged or read anything. We list the corrections, the amendments and the limits.

1. Introduction

Pipelines that combine several language-model outputs into one answer are widely used. Mixture-of-Agents (MoA) [1] reports that a stack of six open proposer models and an aggregator outperforms GPT-4o on AlpacaEval 2.0, and attributes the gain to "collaborativeness": a model shown other models' drafts writes a better response. In advisory settings (healthcare, legal and financial analysis) a common variant gives the proposers specialist roles, on the expectation that specialists contribute knowledge and caution a single generalist would lack.

This project began with one question: does such a council of specialists answer better than one model? On the project's own item sets the question could not be answered. On a 60-item battery with checkable answers a single 20B model answered 57 of 59 items correctly and a specialist-plus-aggregator pipeline 56 of 60; two harder batteries stopped at a ceiling (Section 3.7). The work then turned to narrower questions about what the aggregation step does with what it is given:

1. Does an instruction to the aggregator change what it does, or only the words it uses?
2. What do LLM preference judges reward in the aggregated answer?
3. What happens to a figure when one contribution, or two contributions, supply it?
4. How much of the proposers' qualification reaches the answer?
5. Can a follow-up question to a proposer, or a list of caveats appended to the report, change the outcome?

Experiment 68 then returned to the first question with public questions that leave room to improve and with setups of equal cost (Section 7).

Prior work has answered parts of these at larger scale. Aggregation pipelines have been evaluated on accuracy as well as preference [1, 2], a single deceptive agent has been shown to damage MoA [3], models are known to adopt supplied content over a correct prior [4, 5], LLM judges are known to penalize expressions of uncertainty [6], and summarization is known to drop scope-limiting detail [8, 9]. What this study adds is small-scale, registered, component-level measurement on one pipeline, together with a record of what went wrong in measuring it.

What this draft claims. Statements scoped to one 20B model, in-house scenarios and items, and local LLM judges and readers: an instruction produces the phrase it names (Section 3.1); two local judges prefer the aggregated answer and mark down the instructed phrase (Section 3.2); the aggregator's answer follows a figure stated in the question over a contribution that misquotes it, and with two conflicting contributions it follows the one listed first about three times in four (Sections 3.3 and 3.4); and reader models act on a decisive appended sentence (Section 5.3). On public multiple-choice questions the pipeline is more accurate than one answer, and a vote of four plain answers is as accurate as the pipeline (Section 7). We also describe a design assembled from these observations (Section 4) and report that it showed no difference from the plain pipeline in judge preference, that in its live chain a second source got the right figure used more often without reliably displacing the wrong one and without the conflict reaching the final answer (Section 6), and that its chain was less accurate than the plain pipeline on the public questions (Section 7).

What changed from v0.2. An internal review found errors in the earlier draft that the repository itself exposes. They are listed in Appendix A. The three with the largest effect: the reported share of caveats "carried through prose" (0.174) came from a parsing error and is 0.030 under the paper's own rule; the slope relating caveats in to caveats out was reported with run-level intervals and is close to the ratio of answer length to input length; and the preference shares counted only the minority of pairs each judge decided consistently. The earlier title and abstract described the assembled design as a method that "hardens" MoA. The evidence does not support that description, and it has been removed.

What changed from v0.3. Draft v0.3 announced six registered experiments. They ran between 1 and 5 October 2026 and are reported here; Section 10 gives each one's outcome by its registered rules. Two of them needed labels from the author that could not be provided and were amended before any of their results was read. One was scored a second time after a fault in a text-matching check was repaired. One result is worded more narrowly than its registration allowed. All four changes are in Appendix D, and Appendix B lists the statements of v0.3 that this draft alters.

What changed from v0.4. Draft v0.4 listed experiment 68, the comparison of accuracy at equal cost, as a design. It was given an execution entry on 9 October 2026, ran until 18 October, and is reported in Section 7. Before it ran, a fifth setup was added so that its primary test changes one thing, and one prediction was given a margin (Appendix D). Appendix C lists the statements of v0.4 that this draft alters.

2. Setup and Methods

The system as tested. The aggregator is `gpt-oss:20b`, a 20B reasoning model served locally, with the one-sentence prompt "You are the lead analyst. Several specialists have written the contributions below. Write the final answer to the decision-maker's question." Three proposers supply contributions. In the early experiments (numbered below 30, scored with a keyword counter that the project later abandoned) the proposers were separate fine-tuned 7 to 8B domain models. In the experiments this paper reports, the proposers are *the aggregator model itself* under prompts of the form "You are a healthcare analyst advising a decision-maker…", one each for healthcare, legal and finance; in several experiments one stored contribution per role per scenario is reused across runs. Fine-tuned models appear only in the comparison of Section 3.7 and the screening test of Section 5.1. The pipeline is therefore a role-prompted self-ensemble with one aggregation layer. It is not MoA as published, which uses six different proposers of 70B and above over several layers, and results here should not be read as results about that system.

Comparisons. The *plain pipeline* is three role-prompted contributions and the aggregator prompt above. The *single answer* is the same model answering the scenario with the same prompt minus the sentence that mentions contributions. The pipeline costs four model calls and the single answer one. Comparisons of equal cost (several samples of the single model with a vote [15, 14], and plain samples combined by the same aggregator [2]) were run only in experiment 68, on accuracy (Section 7).

Scenarios and items. Eighteen advisory scenarios were written for the project, spanning healthcare, legal and finance content. Experiments use 6 to 18 of them. Three item sets were written for specific tests: 6 planted disagreements, 8 planted figures and 11 decision questions. A fourth set was built by script for experiments 63 and 67: 61 arithmetic and rule questions over 14 templates whose answers a generator computes, each with one input figure that can be raised or lowered so that the computed answer changes. Scenarios, items, planted facts and the labels the project notebook calls analyst-authored were produced with an AI assistant (see the statement before the references). One public benchmark is used to evaluate the pipeline, in experiment 68 only: the law, health and business questions of MMLU-Pro [25]. Two public datasets with human labels are used only to check judges: FRANK [24] and the SFU Review Corpus annotated for spec-

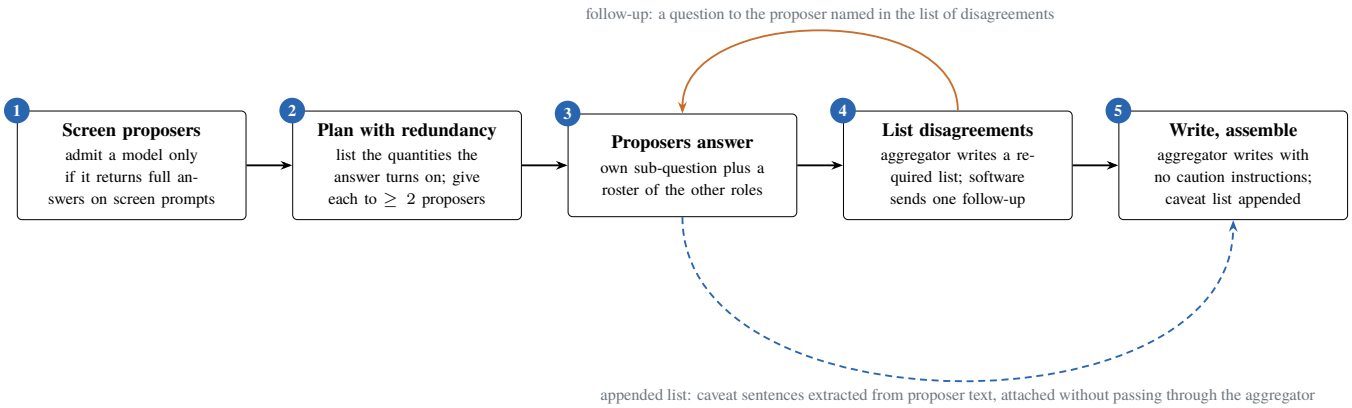


Figure 1: The design described in Section 4. It keeps the proposer and aggregator structure and adds five parts. The dashed blue lane attaches extracted caveats to the final document; the orange loop sends one follow-up question. Section 5 reports which parts have been tested and how.

ulation [26].

Registration. Each experiment has a written entry made before it ran, with predictions, the result that would count as failure, and a computation from existing runs of whether the outcome could occur often enough to measure. There are 66 such entries for the 68 numbered experiments (experiment 68 has a design entry and an execution entry). The entries are in the project's version history ahead of the corresponding results. They are self-registered, and the run records for experiments 25 to 60 were added to the history weeks after their results and carry no timestamps, so the order of registration and data collection cannot be verified from the records. Records made since October 2026 carry time, model digest, seed and settings. The registrations of experiments 62 to 67 were committed before any of their runs and pushed to the public repository 70 minutes before the first run of experiment 63; for experiment 62 the push came 50 minutes after its judge calls had begun. The execution entry of experiment 68 was committed 9 seconds before its first call on a test question and pushed four days later, after its screen had finished. Deviations from the registrations are listed in Appendix D.

Instruments. *Exact phrase*: containment of an instructed phrase. In 2,907 text spans from runs without any instruction, no phrase from the project's registry of instructed phrases occurred. *Sentence labels*: two judges (`gpt-oss:20b` and `qwen2.5:7b`) label each sentence, in batches of ten, for three kinds of qualification (a number labelled as an assumption; different jurisdictions treated separately; a stated condition under which the claim changes), with definitions that name no wording. A sentence counts when both judges flag it. Raw agreement is 0.83 to 0.91, but most sentences carry no qualification: agreement expected by chance is 0.64 to 0.88 and Cohen's kappa [23] is 0.19 to 0.41 (Appendix F). The labels have not been compared with human labels: experiment 62 was registered to do that and could not, because the author was unable to provide the labels (Section 10). Every count that depends on them is provisional and marked †. *Preference*: a judge model sees the question and two answers and names the better one; every pair is judged in both orders. *Exact match* on planted values, after normalisation of number formats. *Reader decision*: a one-word PROCEED or HOLD answer from a reader model. *Computed answer* (experiments 63 and 67): the number on the answer's final line is compared with the answers a generator script computes under the right and under the altered input. *Figure present*: whether a figure's digits occur in a text. As registered, this check does not match a figure typeset with LaTeX thousands separators, which decided one registered check by two items (Section 6). *Option letter* (experiment 68): the letter on the last line that begins with "ANSWER", compared with the benchmark's key; a reply with no such letter, or with two, counts as wrong and is reported.

Preference score. For a pair judged in both orders let $s = 1$ if the first-named side wins both, $s = 0$ if it loses both and $s = \frac{1}{2}$ if the judge's choice follows position. The all-pairs score of a comparison is

$$\text{pref} = \frac{1}{|P|} \sum_{p \in P} s_p, \tag{1}$$

over all pairs P with two parsed judgments. The judges used here choose the first-listed answer in 0.80 to 0.89 of single judgments, so most pairs score $\frac{1}{2}$. The earlier draft reported the share of wins among the pairs with $s \in \{0, 1\}$ only, which discards 63 to 82% of pairs and overstates differences.

Statistics. The unit of analysis is the scenario or the item. For each comparison we average within the unit, take paired differences, and report the mean, a 95% t-interval and the p-value of an exact sign-flip test, with the number of units k. Percentile bootstrap intervals over 6 to 18 clusters, used in the earlier draft, are too narrow [21, 20]. A few intervals are quoted as originally computed and are marked "as first reported". Table 1 lists the five statements we treat as the

paper's main claims, with Holm-adjusted p-values [22]; they were chosen after the results were known, so the adjustment describes rather than protects. The primary predictions of experiments 62 to 68 were named before the runs; Table 9 gives their outcomes by the registered rules. With more than 20 units the sign-flip test uses 20,000 random sign patterns instead of all of them. Everything else is descriptive.

3. What the Plain Pipeline Does

3.1. An Instruction Produces the Phrase It Names

A measurement of an instructed behavior can count compliance with the instruction instead of the behavior. If the instrument detects strings the prompt dictated, the measured quantity M is the sum of compliance C and behavior B,

$$M = C + B, \tag{2}$$

and separating them needs a design in which the dictated string varies while everything else is held fixed. Three arms were run on 18 scenarios with 7 repeats each (378 runs): the aggregator prompt alone as control, and two arms adding the sentence "When you state a number that is an estimate rather than an established fact, label it using the phrase 'modeled at' " (form X) or the same sentence with "taken to be" (form Y). Form X had appeared in the project's prompts for months. Form Y was chosen the day before and appears nowhere in the project's earlier outputs.

Each instructed arm produced its own phrase in about two thirds of answers: $+0.667$ $[+0.450, +0.883]$ for form X and $+0.698$ $[+0.504, +0.893]$ for form Y over control, which produced neither (Figure 2, left). Neither arm produced the other's phrase in 252 runs. A phrase used for months and one chosen the day before behaved alike.

The share of answers containing an assumption-labelled sentence in *other* wording was 0.341 in control, 0.365 under form X (difference $+0.024$ $[-0.080, +0.128]$, $p = 0.75$) and 0.317 under form Y (-0.024 $[-0.131, +0.083]$, $p = 0.81$)†. Answer length was flat across arms. This second result is weaker than the earlier draft said. It is measured with the two-judge labels, which flagged 73 of 110 sentences containing "modeled at" and 35 of 101 containing "taken to be". An instrument that misses a third to two thirds of the instructed phrases could miss an increase expressed in new wording. The supported statement is that the judges saw no change larger than about 0.13; the experiment was registered to detect 0.22 and cannot support a claim of equivalence.

3.2. Local Judges Prefer the Aggregated Answer and Mark Down the Phrase

Three comparisons were judged on the 18 scenarios, 126 pairs each, in both orders. Judge 1 is `gpt-oss:20b`, which wrote both sides of every pair. Judge 2 is a 30B model from another vendor, chosen from three candidates on whether it returned parseable and consistent judgments, without looking at which side it favored.

Pipeline answer against single answer. Judge 1 gave 36 wins, 8 losses and 82 position-driven splits, an all-pairs score of 0.611 ($+0.111$ $[+0.065, +0.157]$ from 0.5, $p = 0.0005$). Judge 2 gave 26, 7 and 93: 0.575 ($+0.075$ $[+0.023, +0.128]$, $p = 0.012$). The two sides' answers differ in length by 1.4%, and restricting to pairs of near-equal length leaves judge 1's share of consistently decided pairs unchanged (0.815 against 0.818). Which property of the pipeline answer the judges respond to is not identified.

Instructed phrase against control. The arms of Section 3.1 differ from control only in the instruction. Judge 1 scored the "modeled at" answers 0.413 (-0.087 $[-0.144, -0.031]$, $p = 0.009$) and the "taken to be" answers 0.440 (-0.058 $[-0.118, +0.003]$, $p = 0.06$). Judge 2 scored them 0.448 ($p = 0.17$) and 0.448 (-0.052 $[-0.092, -0.012]$, $p = 0.023$). Two of the four comparisons reach $p < 0.05$ and all four lean the same way (Figure 3). The registration had predicted the opposite.

This agrees with larger studies: LLM judges including GPT-4o are biased against epistemic markers [6], and human preference data used in alignment are biased against text that expresses uncertainty [7]. Our contribution is a small replication with a counterbalanced phrase on local judges. Read with Section 3.1, it suggests a tension for systems tuned toward such judges: the instruction adds a phrase, and the judges prefer the answer without it.

Judges from other families. Experiment 65 gave the same 378 pairs to four more local judges, under a rule fixed in advance: a judge counts only if it chooses the first-listed answer in fewer than 95% of its single judgments. Two judges of 7 and 8B chose the first-listed answer in all 756 of their judgments, and a third 7B judge in 740. Only `phi4:14b` counted (first-listed in 0.886), the registered test needed two, and its three predictions are not evaluable. Alone, that judge scored the pipeline answer 0.560 ($+0.060$ $[+0.005, +0.114]$, $p = 0.050$; 25 wins, 10 losses, 91 splits) and showed nothing on the two phrase comparisons. On the pairs that it and one of the two earlier judges both decided, they chose the same side in 25 of 29 and 22 of 27. The preference therefore still rests on two judges, one of which wrote the answers. Among the pairs each judge decided, the longer answer won 100 of 131 (judge 1), 70 of 85 (judge 2) and 42 of 70 (`phi4`), and the pipeline answer is the longer one in 72 of 126 pairs; this was computed after the data were seen, and the restriction to near-equal lengths above left judge 1's result unchanged.

Three limits apply. The pipeline answer costs four calls and the single answer one; sampling one model several times and aggregating is known to help [2, 14], so the role structure has not been shown to add anything. On accuracy, the one outcome measured at equal cost, a vote of four plain answers did as well as the pipeline (Section 7). The judges are local models whose single judgments mostly follow position, and at 7 to 8B follow it entirely, so a pool of judges at this scale cannot widen the evidence. And no person judged a pair.

Statement	Experiment	Difference [lo, hi]	k	p	Holm
The instructed phrase "modeled at" appears (share of answers, minus control)	41	$+0.667$ $[+0.450, +0.883]$	18 scenarios	0.0002	0.0012
The primary judge prefers the pipeline answer to a single answer (all-pairs score minus 0.5)	43	$+0.111$ $[+0.065, +0.157]$	18 scenarios	0.0005	0.0020
A reader model acts on a decisive appended sentence (flip share, relevant minus irrelevant)	51	$+0.691$ $[+0.427, +0.955]$	11 items	0.0039	0.0117
The primary judge marks down the answer containing "modeled at" (all-pairs score minus 0.5)	43	-0.087 $[-0.144, -0.031]$	18 scenarios	0.0088	0.0176
A second source lowers how often the first source's figure is stated	47	$+0.450$ $[+0.095, +0.805]$	8 items	0.0312	0.0312

Table 1: The five statements from the experiments before the review that this draft treats as main claims. Differences are means of scenario- or item-level paired differences with 95% t-intervals; p is from an exact sign-flip test; Holm is the adjusted value across the five. The five were selected after the results were known.

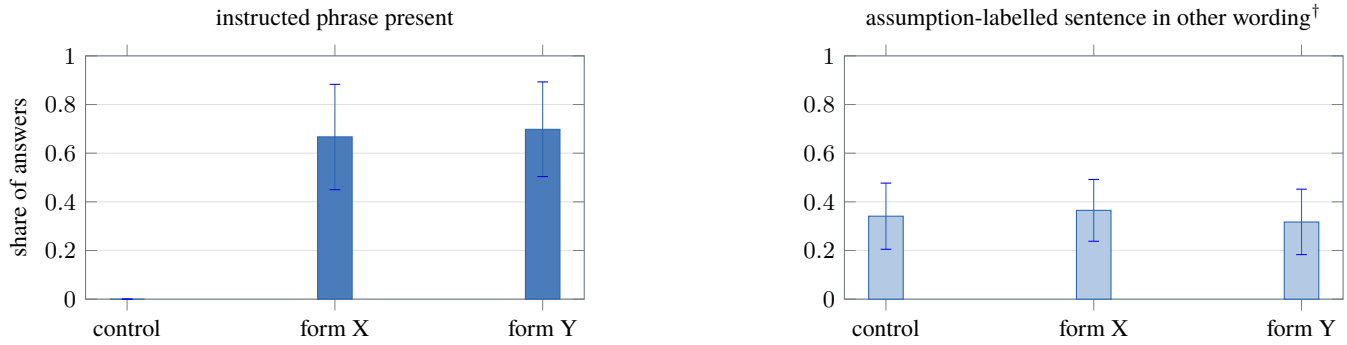


Figure 2: The phrase-swap experiment (378 runs; 95% t-intervals over 18 scenarios). **Left**: each instructed arm produces its own phrase in about two thirds of answers; the control arm produces neither, and neither arm produces the other's phrase ($0/252$). **Right**: the share of answers with an assumption-labelled sentence in other wording does not differ detectably across arms. This panel is scored with the two-judge sentence labels, which find only 0.66 and 0.35 of sentences containing the two instructed phrases, so a change expressed in new wording could be missed.

Item	0	1	2	3	4	5	6	7
one source	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.2
two sources	1.0	0.8	0.0	0.0	0.2	0.6	1.0	0.0

Table 2: Share of five runs in which the planted figure is stated in the answer, per item, with one source and with a second source that gives a different figure (experiment 47).

3.3. With Two Sources for a Figure, the Aggregator States One of Them

Eight scenarios each received one added figure (a price, a rate, a budget) that appears nowhere in the scenario. One proposer's text states a value for it, and the question asks for that figure. In the second arm another proposer states a different value. Answers are scored by exact match.

With one source, the aggregator stated the planted value in 36 of 40 runs. With two, it stated the planted value in 18 of 40 and the second value in 20 of 40. At the item level the drop is $+0.450$ $[+0.095, +0.805]$ ($p = 0.031$, 8 items) and the rise in the second value $+0.500$ $[+0.090, +0.910]$ ($p = 0.063$). The change is all-or-nothing by item (Table 2), and two attempts to predict which source wins, from the model's own plausibility rating and from an elicited map of which role "owns" each quantity, were not evaluable at their sample sizes.

Three points limit the reading. First, repeating the single value is compliance with the question; nothing in the scenario allowed the value to be checked, so the 0.90 rate is not evidence of a failure to verify. That models adopt supplied content readily is established [4, 5]. Second, with two sources the aggregator did not report that two values existed; it stated one. A second source therefore changes the answer in about half of runs without the reader being told there was a choice. Third, when the same design was run with a live planning step and live proposers (experiment 59), the second value was used ($2/40$ against $19/40$, $+0.425$ $[+0.285, +0.565]$, $p = 0.008$) but the drop in the planted value was not established ($25/40$ against $18/40$, $+0.175$ $[-0.069, +0.419]$, $p = 0.22$). The earlier draft described this design as "engineered redundancy" that "halves corrupted-value adoption" and "is causal". Those descriptions are withdrawn. A figure from an early experiment, $0/27$ planted values adopted when a clean source remained, is also withdrawn: in that condition 23 of 27 answers stated neither value, and on several items the planted numbers contradicted the scenario text, which the aggregator followed instead. Section 3.4 repeats the design with a figure the aggregator could check and a scored answer.

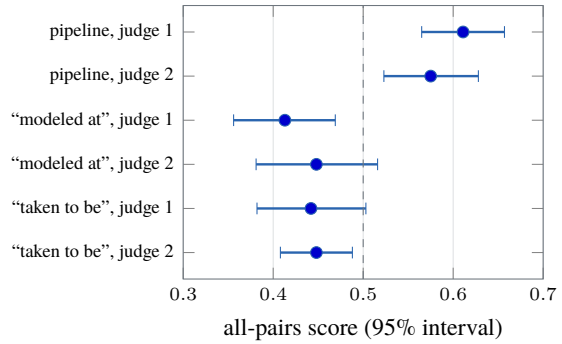


Figure 3: Preference on all pairs (Equation 1; 126 pairs per comparison, both orders; t-intervals over 18 scenarios). Judge 1 is `gpt-oss:20b`, which also wrote every answer; judge 2 is a 30B model from another family. Top two rows: pipeline answer against a single answer. Lower rows: the answer containing the instructed phrase against the control answer. The dashed line is no preference.

3.4. A Figure the Aggregator Could Check

Experiment 63 replaces the planted figure with one that matters to a computed answer. Each of 61 arithmetic and rule questions has one input figure that, raised or lowered by 20 to 25%, changes the answer a generator script computes. The aggregator, with its usual prompt, receives the question and three one-sentence contributions; a contribution that supplies a figure quotes the sentence of the question that contains it. Five layouts differ only in where the right and the wrong figure sit (Table 3). Nothing asks the aggregator to check or to report anything. Each layout was run twice per question; where two contributions carry a figure, the second run swaps their order. The final number was scored by script.

When the question states the figure. With a contribution misquoting a figure that the question itself states, the answer followed the misquote in 0 of 122 runs, against 112 of 122 when the contribution was the only source: by template $+0.923$ $[+0.840, +1.005]$, exact $p = 0.0001$, the smallest value 14 templates allow. This was the registered primary prediction. No reply mentioned the misquote, and the model's output, hidden reasoning included, was no longer than in the layout without a misquote (median 1,006 against 978 tokens). The supported statement is that the answer follows the question's figure. The registration would have allowed “catches a wrong figure”. We do not use that wording: an aggregator that ignored the contributions entirely would produce the same row. The 112 of 122 is close to guaranteed by the layout, in which the question says the figure is given in the contributions.

When two contributions conflict. With the figure removed from the question and two contributions quoting different figures, one right and one wrong, the answer was right in 52 of 122 runs and followed the wrong figure in 62, down from 112 with one source ($+0.416$ $[+0.279, +0.553]$, $p = 0.0002$). Nothing in this layout lets the aggregator tell which figure is right, and the split by listing order shows what it does: the right figure was followed in 42 of 61 runs when its contribution was listed first and in 10 of 61 when it was listed second. Over the two conflict layouts the answer followed the first-listed contribution in 170 of 244 runs and the second in 56; among runs following either, the first-listed share by template is 0.745 [0.664, 0.826]. The drop from one source to two is what choosing one of two produces, and a second source helps only in that the right figure gets a chance of being chosen. The order statistics were computed after the data were seen, and the first-listed contribution is always labelled “Analyst A”, so position and label cannot be separated.

Whether the reader is told. Both conflicting figures appeared in the reply in 41 of the 244 conflict runs (0.186 [0.087, 0.285] by template; registered as below one half). The AI assistant read 60 of these replies to check the count: three of 40 replies without both figures still said that two figures were on offer, and one of 20 with both was a coincidence of digits, which puts the share that let a reader see the conflict near one in five. When the conflict was acknowledged, the stated reason for the choice was the order of mention, caution (“to be conservative”), or an unsupported judgment that one figure was more common, more realistic or a typo.

Two other aggregator models. `phi4:14b` and the 30B model used as judge 2 ran the same 610 prompts with a 4,096-token allowance. Neither followed the misquote (1 and 0 of 122). With conflicting contributions they behaved differently from the main model and from each other. `phi4` showed both figures in 236 of 244 replies and in 62 of them gave an answer for each; the parser, written for single answers, scored 32 of those by their last answer line. The 30B model showed both figures in 164 replies, and 148 of its replies ran to the end of the allowance without a final number. Where one contribution was followed, it was the first-listed one in 0.79 and 0.74 of runs. Whether a reader learns of a conflict thus depended on the aggregator model. That the first-listed contribution usually wins did not.

3.5. Caveats In and Caveats Out

Let s be the number of caveat sentences in the proposers' text and y the number in the answer, both counted with the two-judge labels[†]. Within a scenario set they are related by

$$y = w\,s + c. \tag{3}$$

For `gpt-oss:20b` the slope w is 0.172 and 0.326 in the two scenario sets, both with scenario-level intervals above zero (Figure 4). For a second aggregator model, `phi4:14b`, it is 0.105 and 0.170, and neither interval excludes zero. The earlier draft reported intervals that resampled runs, under which the second model's slope on the newer scenarios appeared to exclude zero; the claim that the slope “replicates for a second writer” is withdrawn.

The earlier draft also read w as a loss: “prose transports 0.11 to 0.33 of supplied qualification.” Table 4 shows what

	gpt-oss:20b				phi4:14b				qwen3-vl:30b			
Question states; contributions quote	right	wrong	other	both	right	wrong	other	both	right	wrong	other	both
right figure; one quotes the right figure	119	0	3	–	91	0	31	–	107	0	15	–
right figure; one quotes a wrong figure	116	0	6	2	96	1	25	2	109	0	13	9
nothing; one quotes a wrong figure	0	112	10	–	0	89	33	–	0	104	18	–
nothing; one wrong, one right	52	62	8	25	44	37	41	116	14	25	83	87
nothing; two different wrong figures	0	112	10	16	1	80	41	120	1	45	76	77

Table 3: Experiment 63: 61 questions with computed answers, five layouts, two runs each (122 runs per layout and model). "Right" and "wrong": the final number equals the answer computed from the right or from a wrong figure. "Other" includes replies with no parseable final number. "Both": the reply contains both of the conflicting figures. The registered tests are those of `gpt-oss:20b`. The other two models fell below the registered check on the first row (0.90 right) and are reported, not tested; 148 of the 30B model's 244 replies in the last two rows were cut at its 4,096-token allowance before a final number.

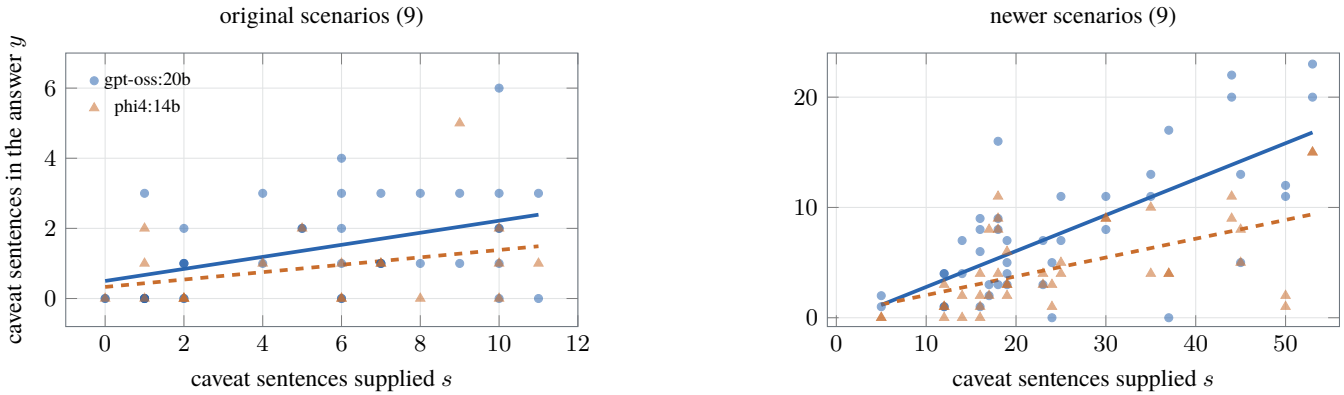


Figure 4: Caveat sentences in the proposers' text against caveat sentences in the aggregator's answer† (144 runs, 18 scenarios, two aggregator models; each point is one run; lines are least-squares fits within each panel). Slopes with intervals over the nine scenarios of each panel: `gpt-oss` 0.172 [+0.067, +0.316] and 0.326 [+0.149, +0.477]; `phi4` 0.105 [−0.013, +0.244] and 0.170 [−0.020, +0.303]. The panels are separate because the two scenario sets supply very different amounts (mean s of 5.4 against 25.3).

Scenarios, aggregator	in	out	length	count
newer, `gpt-oss`	0.183	0.193	0.292	0.307
newer, `phi4`	0.183	0.190	0.178	0.184
original, both	0.133	0.103	0.300	0.231

Table 4: Share of sentences that carry a caveat in the proposers' text (in) and in the answer (out)†; answer length as a fraction of input length; and caveat sentences out as a fraction of caveat sentences in.

the slope mostly reflects. The answer is 18 to 30% as long as its inputs and contains 18 to 31% as many caveat sentences. The share of sentences that carry a caveat is about the same in the answer as in the inputs. A summary is shorter than its sources, and on this evidence caveats are kept roughly in proportion.

Whether a caveat is less likely to be conveyed than an ordinary sentence was the subject of experiment 66. Each of 285 caveat sentences† was matched with an ordinary sentence of similar length from the same proposer text, and each sentence was rated against three stored answers (1,710 sentence–answer pairs): two judges said whether its content is conveyed, fully, partly or not, in the three answer sentences closest to it. The registration required the judges to be checked against 120 pairs labelled blind by the author before the comparison was read. Those labels could not be provided. Before any judge answer was read, the check was replaced by four others with bars fixed in advance (Appendix D): agreement with human labels on 300 summary sentences from the FRANK benchmark [24], and three sets of items with a known answer built from stored material the experiment does not use. The judges passed three. Against the human labels the two-judge rule reached kappa 0.742 and F1 0.906 (bars 0.60 and 0.80); all 50 sentences copied from an answer were rated fully conveyed; and 2 of 50 statements from a different scenario were rated conveyed. They failed the fourth: when a figure was altered in a statement both judges had rated fully conveyed, they still rated it fully conveyed in 5 of 25 cases, against a bar of one in ten. One of the five was a malformed test item and one a statement whose figure was never in the passages; no way of counting reaches the bar.

By the registered rule the comparison is therefore not evaluable. For the record it came out even: 0.696 of caveat sentences and 0.696 of ordinary sentences were rated conveyed at least partly (0.270 and 0.260 fully), a scenario-level difference of +0.004 [−0.128, +0.135] over the 14 scenarios with at least five caveats. That interval is wider than the ±0.10

band registered for calling the two rates equal, so the comparison would not have been evaluable with a passing check either. A judge given the whole answer found the content of about half of the sentences that the three-sentence check had rated absent (13 of 30 caveats, 17 of 30 ordinary sentences). Nothing here suggests that caveats are lost more than other content, and nothing establishes that they are not. Published work finds that summarization and rewriting do lose scope and certainty information [8, 9, 10], so the question remains open here.

On inputs from which every flagged caveat sentence had been removed and the removal confirmed, the aggregator still wrote at least one caveat sentence in 4 of 12 runs†.

3.6. A Follow-Up Reply Is Used When It Is Informative

The aggregator was run in two stages: it first wrote a required list of disagreements among the contributions, then the answer. Six items were built in which two proposers take conflicting positions and a fact neither stated settles the question. Between the stages one of three things was added: nothing; a reply in the right style that restates the proposer's position without the fact; or the same reply with the fact. Two judge models classified which side each answer took (their agreement was 0.70 on the runs analysed and 0.58 overall).

In the runs where the aggregator had named the planted disagreement, it took the fact's side in 7 of 7 runs with the informative reply, 5 of 15 with no reply and 3 of 15 with the uninformative reply. That contrast conditions on naming, which happened in 7, 15 and 15 of 36 runs although the first stage is identical across arms. Over all 36 runs per arm the counts are 15, 10 and 8. At the item level, informative against none is $+0.700$ $[+0.145, +1.255]$ among named runs (5 items) and $+0.139$ $[+0.007, +0.271]$ over all runs (6 items), with exact $p = 0.125$ in both cases; no item goes the other way. With five or six items no contrast here could be strong. We report it as a consistent direction: when the fact arrives, the aggregator uses it. Naming was the scarce step, at 0.21 to 0.58 of runs across three experiments.

3.7. Smaller Observations

Accuracy. On a 60-item battery with checkable answers the single model answered 57 of 59 items correctly and a pipeline of the matching fine-tuned specialist plus the aggregator 56 of 60 (paired difference $[-0.085, +0.034]$, as first reported). Two further batteries with script-computed answers (36 multi-step computations, 30 rule-interaction items) put the single model at 0.977 and 0.983, and were stopped before any pipeline run because there was no headroom. On the one in-house battery where both ran, the difference was indistinguishable from zero and nominally against the pipeline. Experiment 68 repeated the comparison on public questions screened for headroom and at equal cost (Section 7).

Fine-tuned proposers. Two medical fine-tunes of one base model shifted in-domain framework vocabulary by $+0.232$ $[+0.072, +0.412]$ and $+0.109$ $[-0.056, +0.300]$ over the base (as first reported). The intervals overlap almost entirely, so the data do not show that the two differ, as the earlier draft claimed. Stronger evidence for the practical point exists: medical adaptations outperform their base models in 12.1% of comparisons and are worse in 38.2% [12].

Self-assessment. Proposers given a way to defer named the right colleague in 10 of 11 deferrals, and deferred in 0.306 of runs outside their field against 0.083 inside it; the six-item interval for the difference spans zero. A model's plausibility rating predicted which value it would state at 0.544, recorded as not evaluable, and an elicited map of which role owns each quantity at 0.500 $[0.323, 0.739]$ on 28 runs, recorded as a failed prediction; both tests could detect only a value of 0.68 or more. The earlier draft summarized these as self-assessment that "predicts behavior at chance". Three underpowered tests do not support that, and the project's own rule forbids citing an uninformative null as evidence of absence. The question is open; a pre-registered study of small open models finds verbal confidence uninformative for a different reason, saturation at the ceiling [11].

Agreement between proposers. The earlier draft said that claims made by two or more proposers reach the answer no more often than claims made by one. That rested on one of four kinds of qualification in an analysis made with the abandoned keyword counter; the other three kinds showed large positive effects of agreement. The claim is withdrawn. Published work finds that models follow the majority of the evidence they are given [5].

Showing proposers each other's text. With stored sibling contributions made visible, off-domain vocabulary moved by -0.009 $[-0.048, +0.028]$ (as first reported, 18 scenarios), against a registered threshold of $+0.05$. Keeping proposers apart is therefore a cost choice: sibling text adds about 10k input characters per proposer.

4. A Design Assembled from These Observations

The observations above suggest a design with five parts (Figure 1, Figure 5). We describe it as a proposal. Section 5 reports what has been tested, and Table 5 states where the system that was actually assembled differs from the parts that were tested.

Screen proposers. A candidate model is admitted only if every answer on a set of screen prompts reaches 800 visible characters. The earlier draft also listed a probe of domain vocabulary and a format test; neither was validated, and the format test failed to predict anything.

Assign each key quantity to two proposers. A planning step lists the quantities the answer turns on and writes each into at least two proposers' sub-questions. Given Section 3.3, the purpose is to create a visible disagreement when the proposers differ. It does not make the answer immune to a wrong figure: the aggregator follows one of the two, usually the one listed first (Section 3.4), and when this part was tested in the

live chain the disagreement was listed and not passed on (Section 6).

Trigger a follow-up from a required list. The aggregator writes a list of disagreements before it writes the answer. Software reads the list and sends one follow-up question to the first proposer named. A list the aggregator produces whenever asked (391 of 396 stored runs) is a more reliable trigger than a model's report of its own uncertainty, whatever that report is worth.

Append the caveats. Caveat sentences are taken from the proposers' text and attached to the final document as a list the aggregator does not rewrite. The aggregator receives no instruction about caution, since Section 3.1 suggests such an instruction adds a phrase and Section 3.2 that judges mark the phrase down.

Do not tune toward the diagnostics. Phrase counts, the slope w, how often disagreements are named, and judge preference on qualified text are recorded and never optimized (Appendix G).

5. Component Tests

Each part was tested in its own experiment with the failure condition written first. Table 6 collects the results. The rows are separate experiments on separate item sets.

5.1. Screening

Ten local models were screened on two prompts, with the pass or fail recorded before any pipeline run, and each then wrote 18 proposer contributions. The two that failed the screen are the two with the most contributions under 800 characters in the pipeline ($17/18$ and $6/18$; the other eight had 0 or 1 of 18). Under random labelling of two of ten models that has probability $1/45 = 0.022$. The earlier draft reported a pooled contrast with an interval that resampled six scenarios; the unit that matters is the model. The screen and the outcome are the same property, short output, measured on different prompts, so this shows that models which answer very briefly on some prompts do so on others. One model that had passed the screen weeks earlier failed it when repeated, so a stored pass should not be reused.

5.2. Planning Step

Against 41 labels of key quantities written with the AI assistant, the planning step recalled 0.633 $[0.506, 0.758]$ (as first reported), and did better on quantities that were missing from the scenario (0.813) than on stated ones (0.531). Told which quantity mattered, it wrote that quantity into at least two sub-questions in 40 of 40 plans, and proposers passed privately supplied values into their contributions in 0.85 to 0.95 of runs. A first attempt stopped at its registered checkpoint with a useful finding: a proposer conveys a privately supplied fact only when its sub-question concerns that fact (2 of 8 otherwise). The downstream result is the live-chain rows of Table 6.

5.3. Appended Caveat List

What the list holds. The list is built from the proposer sentences that both judges flag†, so it contains all of them by construction; the earlier draft reported this "carriage of 1.000" as a result. Across 119 documents, the aggregator's own prose contains 61 of the $2{,}037$ listed sentence instances word for word (0.030). The earlier draft's 0.174 came from a parsing error: caveats were read back from the list one line at a time, and the quarter of them that span several lines were cut to a first line that is often a single table character. The corrected figure is a verbatim-copy rate. A caveat restated in the aggregator's words counts as missing, so it does not measure how many caveats reach the reader. Experiment 66 attempted that measurement and its judges failed a check (Section 3.5); no validated measure exists.

Judge preference. With the list attached to otherwise identical documents, judge 1 gave 26 wins, 25 losses and 65 splits. No penalty was detected and its absence cannot be claimed.

A reader model uses a decisive sentence. Eleven decision questions were paired with a sentence that should change the decision and a sentence that should not; some sentences argue for holding and some for proceeding. A reader model answered PROCEED or HOLD. With the relevant sentence as the list, reader 1 (`gpt-oss:20b`) changed its decision in 0.727 of reads against 0.036 for the irrelevant sentence: $+0.691$ $[+0.427, +0.955]$, $p = 0.004$ (Figure 6). Placing the same sentence in the prose gave 0.818 (prose minus list $+0.091$ $[-0.061, +0.242]$). Reader 2 (`qwen2.5:7b`) gave $+0.455$ $[+0.104, +0.805]$, $p = 0.063$.

Experiment 64 repeated this with six reader models from five developers (7 to 30B), four of which experiment 51 had not used, on the same 11 items (3,300 reads). Each reader first had to give the item's default answer when nothing was appended, and all six did. Pooled over the four new readers, with experiment 51's prompt and a one-sentence list, the relevant sentence raised the share of changed decisions by $+0.515$ $[+0.277, +0.753]$ ($p = 0.004$, 11 items) over the irrelevant one, which sat at the level of no list for every reader. Removing the clause that told the reader to use anything attached changed uptake by -0.017 $[-0.087, +0.054]$; the registered prediction that the clause mattered failed. Placing the relevant sentence at a random position among nine real caveat sentences lowered uptake by 0.118 $[0.031, 0.206]$ ($p = 0.014$), from 0.55 to 0.43 pooled over the six readers, with most of the loss in the two smallest (Table 7). Position in the list made no difference, and ten real caveats with no relevant sentence did not change decisions (0.116 against 0.103 with no list). The two readers of experiment 51 reproduced their earlier figures in fresh draws. For the smaller readers uptake is all-or-nothing by item, and two of the 11 items moved no new reader.

This removes two limits of experiment 51 (two readers, one of which had written the report while the other fell short of the registered threshold; an instruction naming the attachment) and reduces a third (a one-sentence list). What remains: the

Part	What was tested	What the assembled system runs
Caveat list	sentence-level two-judge labels, a median of 11 caveats per document (experiment 48)	a document-level prompt, at most one quoted sentence per kind per proposer, about 5 per document
Planning step	told which single quantity to cover (experiment 59)	covers its own list of quantities; coverage checked by a rough word match
Follow-up	item-specific questions; the proposer is handed the deciding fact as private notes (experiment 54)	a generic follow-up for the first listed disagreement that names a role; the proposer holds no private information
Screen	800-character screen on ten models (experiment 55)	the same screen on the one model that plays every role
Reply filter	failed twice	off; replies delivered with a flag

Table 5: The assembled system differs from its tested parts.

planted sentence is a decisive fact written for the item, where the system's own lists hold hedges and assumption labels; the control sentence is shorter (165 against 291 characters on average); seven of eighteen items were removed at the pilot check; and every reader is a model. No person read anything.

5.4. Follow-Up with a Live Proposer

Section 3.6 used replies we wrote. Two experiments used a live proposer. In the first the deciding fact sat in the proposer's own first contribution; the aggregator named the disagreement in 0.21 of runs and the experiment stopped at its registered checkpoint. In the second the proposer was given the fact as private notes at reply time. It conveyed the fact in 21 of 21 replies, and the aggregator took the fact's side in 19 of those 21 runs (23 of 36 runs overall). The earlier draft compared this with the no-reply arm of the scripted experiment, stored six days earlier ($5/15$; $10/36$). The aggregator named the disagreement in 0.583 of the new runs against 0.417 of the old ones, so the batches differ, and the item-level contrast has exact $p = 0.125$. We report the counts and claim no effect size. The proposer was handed the fact, so the result shows that the route works when the fact exists in the proposer's notes.

In 4 of the 21 live replies the proposer added invented legal authority to the correct fact, and the aggregator used the reply anyway. A follow-up experiment found invented citations in 0.100 and 0.150 of replies (120 replies; all on one item, where the same two invented decisions recurred). A blocklist of those names caught every case in 141 stored replies, which shows only that the filter runs as written, since its names were taken from those replies.

In 54 runs of the assembled system on the scenarios themselves, with no planted disagreement, the list named a role and a follow-up went out in 46 (0.85).

5.5. Parts That Failed or Were Dropped

A filter meant to drop replies that add nothing failed its registered criteria twice: a two-judge version dropped 3 of 6 uninformative replies, and a stricter version dropped all six but also rejected 52% of genuine replies. Replies are delivered with a flag instead. The ownership map and the plausibility rating (Section 3.7) were not adopted. The rule that proposers do not see each other's text lost its original rationale (Section 3.7) and stands on cost.

6. The Assembled System Against the Plain Pipeline

The assembled system was run three times on each of the 18 scenarios and its documents were paired with stored answers from the plain pipeline (108 pairs, both orders, two judges). On all pairs judge 1 scored the assembled system 0.528 ($+0.031$ [-0.069, $+0.132$], $p = 0.57$) and judge 2 scored it 0.477 (-0.023 [-0.060, $+0.014$], $p = 0.31$). No difference was detected.

Four things limit this result. No equivalence margin was registered, so it does not show that the two are equally preferred. Judge 2 decided only 15 pairs consistently (5 for the assembled system, 10 against), in 9 of 18 scenarios, and its earlier reported interval of $[0.083, 0.562]$ does not exclude a large loss; the earlier draft's statement that large effects were excluded in both directions holds for judge 1 only. The plain pipeline's answers had been stored three weeks earlier. And the comparison is a bundle: several parts change at once.

Cost. One question takes about fourteen model calls in the assembled system against four in the plain pipeline, and 14 to 18 minutes on the hardware used. The earlier draft's phrase "at no cost" is withdrawn.

A wrong figure in the live chain. Experiment 67 put the 61 questions of Section 3.4 through the chain with one model in every role: a planning step, three role-prompted proposers, the required list of disagreements, one follow-up and the final answer. The figure was removed from the question and reached the chain only through a proposer's private notes: the wrong figure in one proposer's notes and, in the second arm, the right figure in another's. The plan and the texts of the two proposers whose notes do not change were generated once and shared between the arms (122 runs, 710 model calls).

The registration required each figure to appear in its proposer's text in at least 0.75 of items before the predictions could be read. As first scored, the wrong figure appeared in 44 of 61 texts and the check failed. Both missing texts contain the figure typeset with LaTeX separators, which the registered text-matching check does not recognise; the fault had been recorded three days earlier, in the reading of experiment 63, and not repaired. At the author's decision the experiment was scored again with that one repair, giving 46 of 61 and, for the right figure, 51 of 61. Both scorings are on record

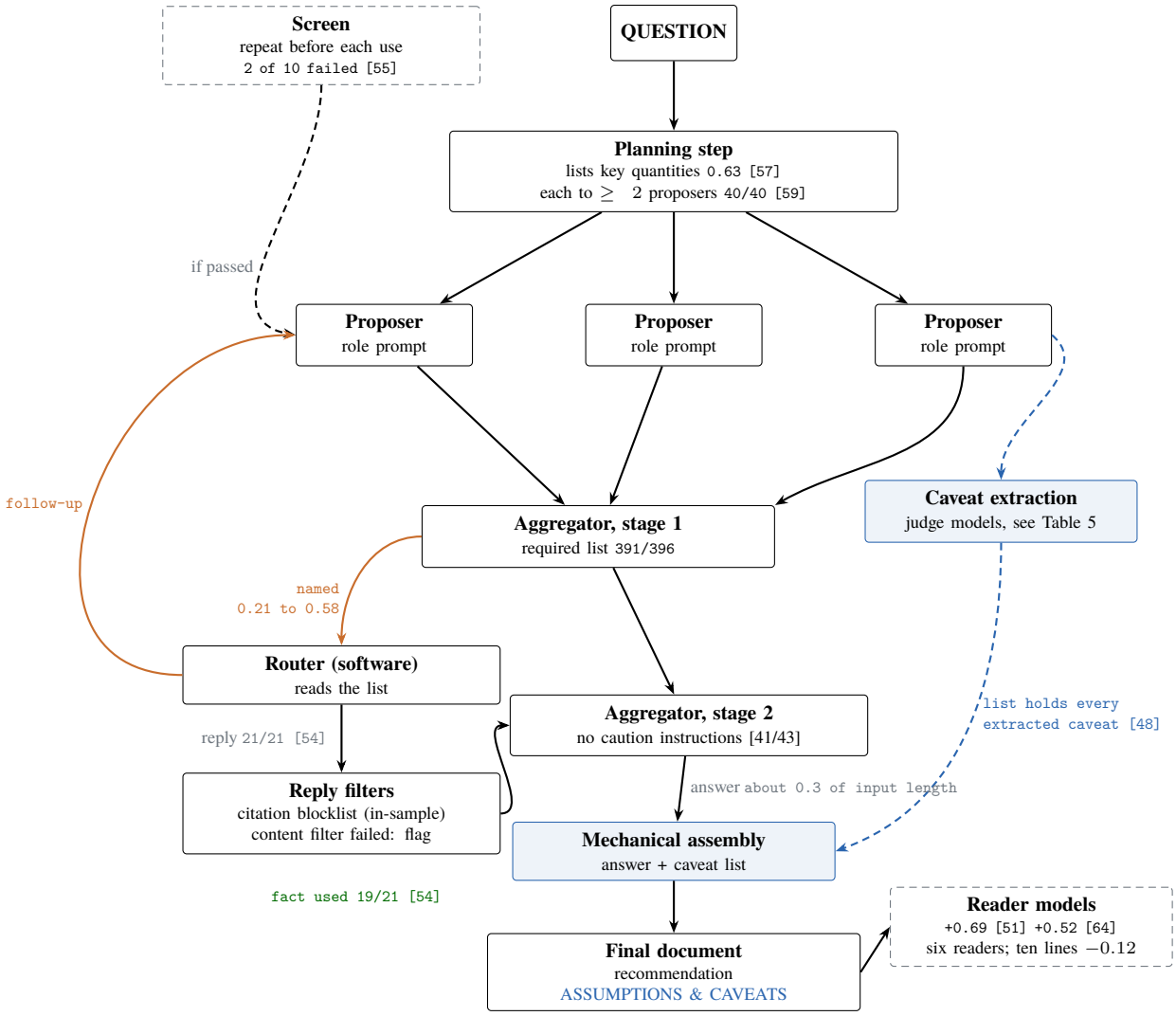


Figure 5: The design on one question. Labels give the measurement and the experiment number in brackets. Dashed blue: the caveat list, which the aggregator does not rewrite. Orange: the follow-up loop. The measurements are from separate experiments on separate parts (Section 5); they are not outcomes of the assembled system, which has been evaluated on judge preference and, for its live chain, on a planted wrong figure (Section 6) and on accuracy (Section 7).

(Appendix D). The check then passes by one item, and what follows should be read with that history.

With a second source, right final answers rose from 2 of 61 to 19 of 61 ($+0.290$ $[+0.188, +0.393]$, $p = 0.0002$, 14 templates). The share following the wrong figure fell from 31 to 24 of 61 ($+0.130$ $[+0.010, +0.250]$); the exact test gives $p = 0.055$, so the registered primary prediction, which required $p < 0.05$, was not met. This is the pattern of experiment 59 on 61 scored items: the second source's figure is used, and the fall in the wrong figure is not established. The final answer showed both figures in 2 of 61 runs, although the aggregator's own list of disagreements named both in 31. The step meant to expose a disagreement recorded it and did not pass it on.

Three further observations were not registered tests. Of the 43 answers that followed one of the two proposers, 35 followed the one whose contribution was listed first: the list of disagreements names that proposer first, the follow-up goes there (to the holder of the wrong figure in 22 of 33 runs when it was listed first; to the holder of the right figure in 16 of 28 when that one was), and the answer keeps that proposer's figure. When the figure reached no proposer's text (21 runs), the chain still returned a number every time, and in 20 of them it was neither the right nor the planted answer. And the live chain showed the conflict in fewer final answers than the aggregator alone did in Section 3.4 (0.03 against 0.20).

What has not been measured. The assembled system's effect on caveats reaching a reader has not been measured as a system, and its appended list and its screen played no part in experiments 67 and 68.

Test (experiment)	Counts, comparison → treatment	Difference [lo, hi]	p	k
Second source, planted: first figure stated (47)	$36/40 \to 18/40$	$+0.450$ $[+0.095, +0.805]$ (drop)	0.031	8 items
Second source, planted: second figure stated (47)	$0/40 \to 20/40$	$+0.500$ $[+0.090, +0.910]$	0.063	8 items
Second source, live chain: first figure stated (59)	$25/40 \to 18/40$	$+0.175$ $[-0.069, +0.419]$ (drop)	0.22	8 items
Second source, live chain: second figure stated (59)	$2/40 \to 19/40$	$+0.425$ $[+0.285, +0.565]$	0.008	8 items
Misquoted figure followed: only source → question states the figure (63)	$112/122 \to 0/122$	$+0.923$ $[+0.840, +1.005]$ (drop)	0.0001	14 templates
Wrong figure followed: one contribution → a second with the right figure (63)	$112/122 \to 62/122$	$+0.416$ $[+0.279, +0.553]$ (drop)	0.0002	14 templates
Live chain, computed answers: wrong figure followed (67, re-scored)	$31/61 \to 24/61$	$+0.130$ $[+0.010, +0.250]$ (drop)	0.055	14 templates
Live chain, computed answers: right answer (67, re-scored)	$2/61 \to 19/61$	$+0.290$ $[+0.188, +0.393]$	0.0002	14 templates
Follow-up, scripted reply, runs that named the disagreement (44)	$5/15 \to 7/7$	$+0.700$ $[+0.145, +1.255]$	0.125	5 items
Follow-up, scripted reply, all runs (44)	$10/36 \to 15/36$	$+0.139$ $[+0.007, +0.271]$	0.125	6 items
Follow-up, live proposer, named runs; control stored six days earlier (54)	$5/15 \to 19/21$	$+0.583$ $[+0.121, +1.046]$	0.125	5 items
Reader acts on an appended sentence, reader 1 (51)	$0.036 \to 0.727$	$+0.691$ $[+0.427, +0.955]$	0.004	11 items
Reader acts on an appended sentence, reader 2 (51)	$0.091 \to 0.545$	$+0.455$ $[+0.104, +0.805]$	0.063	11 items
Reader acts on an appended sentence, four new readers pooled (64)	$0.11 \to 0.63$	$+0.515$ $[+0.277, +0.753]$	0.004	11 items
Uptake with the instruction clause minus without it, six readers (64)	0.534 against 0.551	-0.017 $[-0.087, +0.054]$	0.65	11 items
Uptake in a list of ten minus alone, six readers (64)	0.432 against 0.551	-0.118 $[-0.206, -0.031]$	0.014	11 items
Caveat minus matched ordinary sentence, rated conveyed (66; judges failed a check)	0.696 against 0.696	$+0.004$ $[-0.128, +0.135]$	0.95	14 scenarios
Document with caveat list against without, judge preference (48)	26 wins, 25 losses, 65 splits	$+0.002$ $[-0.120, +0.124]$	1.0	17 scenarios
Screen: the two models that fail are the two with most short outputs (55)	2 of 10 fail	exact, under random labelling	0.022	10 models
Assembled system against plain pipeline, preference, judge 1 (61)	27 wins, 21 losses, 59 splits	$+0.031$ $[-0.069, +0.132]$	0.57	18 scenarios
Assembled system against plain pipeline, preference, judge 2 (61)	5 wins, 10 losses, 93 splits	-0.023 $[-0.060, +0.014]$	0.31	18 scenarios

Table 6: Component tests. Differences are means of item- or scenario-level paired differences with 95% t-intervals and exact sign-flip p-values. For preference rows the difference is the all-pairs score minus 0.5.

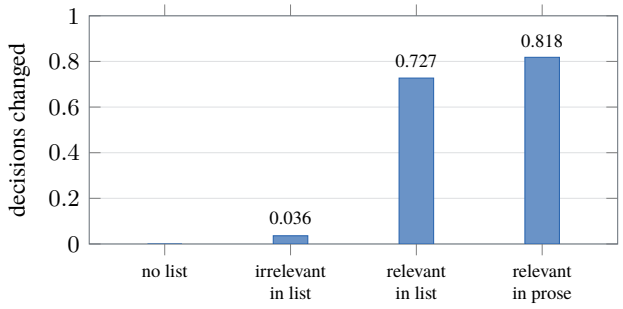


Figure 6: Share of a reader model's one-word decisions changed by a planted decision-relevant sentence, by where it is placed (reader 1, 11 items, 55 reads per bar). The list in this experiment is one sentence long and the reader is told to use "anything attached".

	irrelevant	relevant sentence in a list of		
Reader	1	1	5	10
`gpt-oss:20b`	0.09	0.85	0.76	0.75
`phi4:14b`	0.11	0.85	0.80	0.69
`qwen3-vl:30b`	0.09	0.64	0.65	0.65
`qwen2.5:7b`	0.16	0.55	0.53	0.49
`mistral:7b`	0.09	0.56	0.40	0.36
`llama3:8b`	0.13	0.53	0.29	0.36

Table 7: Experiment 64: share of one-word decisions changed, by reader and list length, with the prompt that does not mention attachments (55 reads per cell). With no list the share is 0.09 to 0.15.

7. Accuracy at Equal Cost on Public Questions

The project's first question, whether a council of specialists answers better than one model, had stopped at a ceiling on its own item sets (Section 3.7), and every preference result compared four calls with one. Experiment 68 was registered as a design on 1 October 2026 to address both, and given an execution entry on 9 October that fixed the items, the screen, the setups and the rules before any test question was sent to a model.

Questions. The law, health and business questions of the MMLU-Pro test split [25]: 2,708 multiple-choice questions with up to ten options and a fixed key. **Screen.** In an order fixed by a hash of each question's identifier, the single model answered each candidate up to six times with the direct-answer prompt, and a question was kept if it was right in two

Setup	calls	tokens	letter	right
(a) one direct answer	1	1,960	0.990	0.496
(b) vote of four direct answers	4	7,839	1.000	0.561
(b2) three direct + aggregator	4	7,421	0.993	0.544
(c) three role-prompted + aggregator	4	7,936	0.988	0.566
(d) assembled chain	7–12	23,204	0.946	0.456

Table 8: Experiment 68: 136 MMLU-Pro questions, three replicates of every setup, `gpt-oss:20b` in every part. Setup (c) is the plain pipeline; (b2) is called "plain drafts" in the text. "Tokens": mean prompt and output tokens per run. "Letter": share of runs that gave a readable option letter. "Right": accuracy, the mean over questions of each question's mean over replicates. Setup (a) is scored on all four direct draws of each replicate (1,632 runs); the others on 408 runs each. No run hit the context limit. The chain made a median of 8 calls.

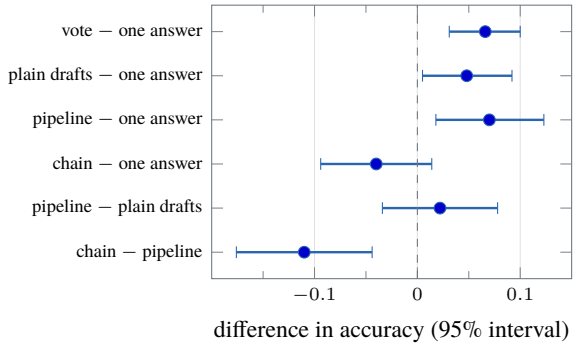


Figure 7: Experiment 68, question-level paired differences with t-intervals over 136 questions. "Plain drafts" is setup (b2): three direct answers given to the same aggregator as the pipeline. The second row from the bottom is the registered primary test (P68.2); the bottom row is P68.3, whose registered margin was -0.05.

to four of the six. Draws stopped once that decision was fixed, which gives the same decision as six draws in every possible sequence. Each section stopped at 50 kept questions or 400 candidates. The screen kept 136 questions: law 50 of 193 candidates, health 50 of 381 and business 36 of 400, where the model's screen draws were right in 0.42, 0.75 and 0.84 of cases. The screen never saw a pipeline answer, and its draws were not reused.

Setups. Five setups answered every kept question three times in one session, with `gpt-oss:20b` in every part (Table 8). Setups (a), (b) and (b2) share the four direct draws of each replicate: (a) is their mean accuracy, (b) the plurality of their letters, and (b2) gives three of them to the aggregator. Setup (c) gives the aggregator three answers written under the healthcare, legal and finance role prompts of Section 2 instead. Setups (b2) and (c) differ only in the proposers' system prompt: the user message (the question, its options and a request to end with an answer line), temperature, token allowance, aggregator prompt and the format of the contributions are the same, and (b), (b2) and (c) each make four calls with token counts within 7% of each other. So that the aggregator prompt is true of both setups, it says three analysts wrote the contributions, labels them A to C in a shuffled order and asks for an answer line; otherwise it is the prompt of Section 2. Setup (d) is the chain of Section 4: a planning step, three proposers with sub-questions and a roster, the required list of disagreements, one routed follow-up with the reply filter, and the aggregator's final step. It differs from the design in three ways: the screen was not repeated before the run, no caveat list was built (it is attached after the answer and could not change it), and the final prompt adds the request to answer exactly and end with an answer line, as setups (a) to (c) do.

Predictions. P68.1: the pipeline (c) is more accurate than one answer (a). P68.2, the primary: the pipeline (c) is more accurate than the plain drafts (b2), the test of whether the role prompts add anything beyond sampling. P68.3: the chain (d) is no more than 0.05 less accurate than the pipeline (c). A prediction is supported when its question-level interval lies above zero (-0.05 for P68.3), fails when it lies below zero, and is otherwise not shown. Computed before the runs, 150 questions and three replicates give a half-width of about 0.07, so a difference of 0.05 would usually not be shown.

Results. One answer was right in 0.496 of runs, which shows that the screen left room on both sides. The pipeline was right in 0.566: P68.1 is supported ($+0.070$ $[+0.018, +0.123]$, $p = 0.009$). Against the plain drafts given the same aggregator it led by $+0.022$ $[-0.034, +0.078]$ ($p = 0.49$), so P68.2 is not shown. The interval leaves room for a gain of up to about 8 points from the role prompts, or a loss of 3; it is a limit of size, not evidence that they add nothing. The chain was right in 0.456: P68.3 fails (-0.110 $[-0.176, -0.044]$, $p = 0.002$), and against one answer it was -0.040 $[-0.094, +0.014]$ (Figure 7).

Where the gain over one answer comes from. The rest of this section was computed after the results were seen. A vote of four plain answers gained $+0.066$ $[+0.031, +0.100]$ over one answer, and the pipeline minus the vote, the comparison the design first registered, is $+0.005$ $[-0.061, +0.071]$. A role-prompted draft is no more accurate than a plain one (-0.008 $[-0.041, +0.025]$), and the role that matches the question's section was not consistently the most accurate (in business it was the least). The aggregator mostly follows the majority of its three drafts. It gave a letter none of them gave in 3 to 4% of replicates, overruled three unanimous drafts in 5 of 408 in each setup, and on two-to-one splits took the minority letter in about one case in seven, moving away from the right answer as often as toward it. Its accuracy against the majority of its own drafts was 0.544 against 0.547 for plain drafts and 0.566 against 0.529 for role-prompted drafts. A plausible reason for the difference, not tested, is that role-prompted drafts carry more reasoning: one plain draft in three is a bare answer line, against one role-prompted draft in seven. The final letter matched the draft listed first, second and third about equally often (0.65, 0.66 and 0.64 for plain drafts), so the first-listed preference of Section 3.4, where contributions conflicted on a

figure, does not appear when they disagree on a choice.

The chain. The chain gave no readable letter in 5.4% of runs against 1.2% for the pipeline, but this does not explain its result: among runs that gave a letter it was right in 0.482 against 0.573, and a lenient reading of its unreadable replies moves the difference to -0.105. Its errors also differ in kind. Of its wrong letters, 32% are the letter the model's direct answers chose most often for that question, against about 50% for every other setup: the chain's errors are less often the model's own usual wrong answer. The follow-up went to the healthcare proposer in 233 of 408 runs, including 86 of 150 on law questions, because the router takes the first role named in the list. Which part of the chain costs accuracy is not identified: the comparison changes several parts at once, and the experiment stored the chain's final reply but not its plan, contributions or list. The chain was built for open advisory questions with a written report; these are knowledge questions with one key, which may be the hardest case for dividing a question into three sub-questions. The result says nothing about its reports.

Checks and limits. After the screen and before any setup's result was read, five kept questions were found to owe their place to replies that gave the right letter in a form the registered parser does not read (a bare letter, or a letter inside a formula); without them the three registered comparisons and the comparisons with the vote move by at most 0.002. The questions are public and may be in the model's training data. A screen for questions the model is unsure about also selects questions whose key is disputed or wrong; such questions add noise to every setup alike. Two of the three role prompts are off-topic on every question, as the design intends. Business has 36 questions and its section-level intervals are wide. And all of it is one model.

8. Limitations

One model in every role. Proposers and aggregator are the same 20B model in the experiments that carry the paper's claims. Results may not hold for different or larger models, or for proposers that differ from the aggregator.

The baseline is not the published method, and cost is matched only for accuracy. See Section 2. Every result on preference, figures and caveats compares four calls with one; only experiment 68 compares setups of equal cost.

No human measurement. Every judge is a local model of 7 to 30B and every reader is a model. No sentence label, preference or decision comes from a person. Experiments 62 and 66 were registered to obtain labels from the author; the labels could not be provided. Public human labels on neighbouring tasks were used instead, as a check with registered bars for one experiment and as a reported figure for the other. The items and several label sets were written with the same AI assistant that designed the experiments, which is the instrument-independence problem of Equation 2 at one remove.

The sentence instrument is not validated. Its chance-corrected agreement is low, it misses a third to two thirds of the instructed phrases, a judge shown the same sentence in a different batch agrees with its own earlier label at kappa 0.42 to 0.71, and the registered comparison with a person could not be made (Appendix F). Results marked † depend on it and stay provisional.

The conveyed judge failed a check. The two-judge rating of whether a sentence's content reaches the answer agreed with public human labels and failed on statements with an altered figure. The comparison of caveats with ordinary sentences is not evaluable (Section 3.5).

Few units. Claims rest on 5 to 18 items or scenarios written in-house, on 14 templates of scripted questions, or, for accuracy, on 136 public questions in which differences under about 6 points cannot be seen. Several effects are all-or-nothing by item. The results on figures that can be checked come from arithmetic and rule questions with one-sentence contributions, not from the advisory scenarios.

Constructed outcomes. In several component tests the outcome was close to guaranteed by the design: a question that asks for the planted figure; a list scored against the sentences it was built from; a decisive sentence placed before the question; a proposer handed the deciding fact; a question that points to the contributions for a figure only they contain. These tests show that a route works. They do not show how the pipeline behaves on inputs nobody arranged. Experiment 68 is the exception: its questions were written by others and chosen only for being uncertain for the single model.

Registration. The registrations are the author's own, in the author's repository, and until October 2026 the run records could not show when they were made. Deviations occurred, among them two registrations amended before their data were read, one experiment scored a second time, and one design changed in its execution entry before it ran (Appendix D).

Many tests. About ninety registered predictions were made over the program, of which about 35 were recorded as failed. The five claims of Table 1 were selected afterwards. The seven experiments added since draft v0.3 made 29 more: 11 were supported or passed (three of them only after the second scoring of experiment 67), 3 failed, 1 check failed, 7 could not be evaluated, 1 was not shown and 6 were reported without a test.

Error rate of the record. The project notebook logs four errors in code that printed verdicts, three text-matching errors, five token-budget failures, one illustrative quotation in an early draft that was not real data, and one experiment run on the wrong condition. The review behind draft v0.3 found two more defects in headline numbers. In the experiments added since draft v0.3, a text-matching check missed typeset figures and decided one registered check by two items, the answer parser mis-scored replies that gave two answers, and a set of test items built to check a judge contained a malformed item. In experiment 68 the letter parser did not read a bare letter or a letter inside a formula, which kept five easy questions in the screen, and the chain's intermediate texts were not stored,

so the part that cost accuracy cannot be traced. Others may remain.

9. Related Work

Aggregation pipelines. MoA [1] reports preference-judged results on AlpacaEval, MT-Bench and FLASK and, in an appendix, accuracy on MATH by layer. Self-MoA [2] shows that aggregating repeated samples of the single best model outperforms mixing different models, on AlpacaEval and on MMLU, CRUX and MATH, and sampling one model many times with a vote scales with the number of samples [14, 15]. These are the missing baselines of Section 3.2; experiment 68 ran both at equal cost on accuracy, and its result, that role prompts did not separate from plain samples of one model, is in line with Self-MoA's (Section 7). A single strongly prompted agent can match multi-agent discussion [19]. The earlier draft said that every published MoA number came from a preference judge; that was wrong.

Wrong inputs. Wolf et al. [3] insert one deceptive agent into a three-layer MoA and measure the damage on AlpacaEval 2.0 (win rate 49.2% to 37.9%) and on QuALITY accuracy, with voting-style defenses that recover most of it. Models adopt incorrect supplied content over a correct prior more than 60% of the time [4] and are highly receptive to coherent external evidence [5]. The earlier draft said no published evaluation measures whether a corrupted proposer corrupts the answer; that was also wrong. Sections 3.3 and 3.4 are small numeric-value variants, with a second-source control and with a control in which the question itself states the figure.

Judges and uncertainty. LLM judges including GPT-4o are biased against epistemic markers [6]; human preference data are biased against text that expresses uncertainty [7]; judges show position bias that varies by model [16], length correlations under preference optimization [17], and sycophancy [18]. Our judges' position bias should be read as a property of the small local judges tested.

Loss of qualification. LLM summaries overgeneralize scientific findings in 26 to 73% of cases [8], rewriting raises expressed certainty more often than it lowers it [9], and handoff summaries between agents keep facts while weakening the conditions on their use [10]. Section 3.5 attempted a comparable measurement and could not validate its judges.

Specialist models and confidence. Medical adaptations rarely outperform their base models [12]. Verbal confidence in small open models fails a pre-registered validity screen [11].

Registration. Pre-registration of experiments with AI agents has a proposed template that lists outcome-contingent redesign among the main risks [13]; the deviations in Appendix D are instances. We are not aware of an earlier experiment that swaps the phrase named by an instruction under otherwise identical prompts, or of one that tests an appended caveat list with a matched irrelevant control.

10. The Seven Experiments Registered After the Review

Six experiments were registered on 1 October 2026, each redesigned for one person on one machine and each addressing a limitation of draft v0.3. They ran back to back between 1 and 5 October in about 93 hours of machine time. A seventh, registered the same day as a design, was given an execution entry on 9 October and ran from 9 to 18 October in about 212 hours. Table 9 gives each outcome by the rules written before the runs.

Two of the first six added to what the paper claims (63 and 64). One qualified the assembled design (67). Three could not answer their question (62, 65 and 66), and in each case the reason is a property of the measuring tools available to one person with local models: no human labels, judges of 7 to 8B that choose by position, and a conveyed judge that misses an altered figure one time in five. The seventh (68) gave the project's first question its first test with room to improve and equal cost, for one model: more accurate than one answer, not distinguishable from sampling the model several times, and the assembled chain was less accurate.

What remains open is therefore much of what draft v0.3 listed: a human check of the sentence judges and of the conveyed judge; preference judges of 14B and above from other model families, and human judges; human readers; an aggregator other than `gpt-oss:20b` that passes the clean-layout check of experiment 63; whether role prompts add a few points of accuracy, which would need about four times the replicates or questions of experiment 68; which part of the assembled chain costs accuracy; whether the accuracy results hold for another model or for open questions with a checkable answer; a second annotator, an independent rerun of the analysis and a much larger model.

11. Conclusion

On one 20B model, eighteen in-house scenarios and a scripted set of 61 questions with computed answers, these behaviors of a role-prompted aggregation pipeline were observed under registered tests. An instruction to mark estimates produces the phrase it names. Two local judges prefer the aggregated answer to a single answer and mark the instructed phrase down. The aggregator's answer follows a figure stated in the question over a contribution that misquotes it, without mentioning the misquote. Given two contributions that conflict, it follows one of them, usually the first listed, and the model studied most rarely says there were two. Reader models from five developers act on a decisive sentence appended to the report. And on 136 public questions that leave room to improve, the pipeline is more accurate than one answer, but no more accurate than a vote of four plain answers at the same number of calls.

A design assembled from these observations was indistinguishable from the plain pipeline on a weak preference test, and on the public questions its chain was 11 points less accu-

No.	Question	Outcome by the registered rules
62	Do the sentence judges agree with a person?	**Not evaluable.** The author's blind labels, the registered reference, could not be provided. Reported instead: agreement between pairs of six judges and of each judge with itself (Appendix F). Counts marked † stay provisional.
63	Does the aggregator's answer follow a wrong figure it could check?	**Check passed, four predictions supported.** Primary: the misquoted figure is followed in $0/122$ runs when the question states the right one, against $112/122$ when it is the only source ($+0.923$, $p = 0.0001$). Two other aggregator models failed their own check and are reported (Section 3.4).
64	Do other reader models act on an appended caveat, without the instruction, in a long list?	**Primary supported** on four new readers ($+0.515$, $p = 0.004$). The prediction that the instruction matters **failed** (-0.017). The prediction that a list of ten dilutes was **supported** (-0.118, $p = 0.014$) (Section 5.3).
65	Does the judge preference repeat with judges from other families?	**Not evaluable.** Three of four added judges chose the first-listed answer in at least 98% of judgments and did not count; the fourth leaned the same way as the earlier judges ($+0.060$) (Section 3.2).
66	Are caveats conveyed less often than ordinary sentences?	**Not evaluable.** After an amendment, the judges passed three of four checks and failed the fourth. The rates were equal (0.696 each), with an interval wider than the registered band (Section 3.5).
67	Does a second source help in the assembled chain?	First scored: **check failed** ($44/61$), predictions not evaluable. Re-scored after a repair to the text-matching check: check passes ($46/61$); primary **not met** (wrong figure followed $31/61$ against $24/61$, $p = 0.055$); the second source's figure is used (**supported**, $+0.290$); the conflict is in $2/61$ final answers (**supported**) (Section 6).
68	Is the pipeline more accurate than the same model sampled four times?	On 136 public questions with room to improve: the pipeline beats one answer (**supported**, $+0.070$, $p = 0.009$); against three plain drafts given the same aggregator, the primary test, **not shown** ($+0.022$ [-0.034, $+0.078$]); a vote of four plain answers did as well as the pipeline; the assembled chain fell more than the registered margin below the pipeline (**failed**, -0.110, $p = 0.002$) (Section 7).

Table 9: The experiments registered on 1 October 2026 and their outcomes. The primary predictions were named in the registrations. Experiments 62 and 66 were amended on 2 October, before any of their judge answers was read; experiment 67 was scored a second time on 5 October; experiment 68's execution entry of 9 October added a setup and changed its primary prediction before any run (Appendix D).

rate than the plain pipeline. In its live chain a second source got the right figure used more often, and the step meant to expose disagreements listed them without passing them on. Three questions the review raised could not be answered by one person with local models: whether the sentence judges match a person, whether caveats are conveyed less often than other content, and whether the judges' preference extends beyond two models. Whether any of this helps a person making a decision is unknown.

The measurement lessons may be the more durable result. An instrument that counts strings the prompt dictated measures compliance. Raw agreement between judges says little when one answer dominates. Intervals over a handful of scenarios need methods built for few clusters. A text-matching check has to be tried on the writing habits of the model whose text it will read. Small judges given two long answers choose the first. A pipeline compared with one call should also be compared with the same number of plain calls. And a number that lands inside the expected range still needs checking.

Use of AI Assistance

An AI assistant (Claude, Anthropic) was used throughout this project. It proposed and drafted most of the registrations and the experiment code, wrote the scenarios, test items, planted facts and the labels the project notebook calls analyst-authored, ran the analyses, conducted the three internal audits and the October 2026 review, ran and monitored the seven experiments that followed, wrote the readings of their results, proposed the two amendments, the checks that replaced the author's labels and the changes to experiment 68's design, and drafted this text. The author directed the work, decided what was run and what is claimed, decided the amendments, the second scoring of experiment 67, the wording of the result of experiment 63 and the download of the public questions for experiment 68, and takes responsibility for the content. Because one source wrote the items and labelled several outcomes, results that depend on those labels should be read with that in mind.

Data and Code

All registrations, run records, analysis code and the project notebook are in the project repository. The corrected figures of draft v0.3 are produced by `train/run_stage0_corrections.py` (output in `bench/analysis/stage0/`) at commit `bb7fcab`. The review and its reproduction script are `docs/BOARD_REVIEW_2026-10-01.md` and `bench/analysis/board_review/checks.py`. Experiments 62 to 67 are registered at commit `6402fc0` and the two amendments at `29f491f`; experiment 68's execution entry is at `ddabf2a` and its runner is `train/run_cell68_equalcost.py`. Each of their run records carries its registration commit, the hash of the script that made it, the model digest, the seed and the time. Figures quoted beyond the registered outputs come from the check scripts in each experiment's analysis directory, and the second scoring of experiment 67 from `bench/analysis/cell67/rescore.py`. The sources, licences and checksums of the three pub-

lic datasets are in `docs/EXTERNAL_DATA.md`. None of these scripts calls a model.

Impact Statement

This work concerns language-model pipelines in advisory settings. Its findings are limited to 7 to 30B local models, English advisory scenarios written for the project, and small numbers of items, and they involve no human participants. They should not be used to justify deploying such a pipeline for real advice. The observation that an aggregator given two different figures follows the one listed first, and that the model studied most does so without flagging the conflict, is a reason for caution wherever such pipelines are used. So is the finding that, on the questions tested, a fuller pipeline was less accurate than a simple one, and a plain vote matched the role-prompted pipeline.

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A. Corrections from Draft v0.2

Each item gives the earlier statement and what the repository shows.

1. Caveats carried through prose. Earlier: the appendix "carries 1.000 of proposer caveats against 0.174 through prose." The 0.174 was a parsing error (Section 5.3); the verbatim rate is $61/2{,}037 = 0.030$, and 1.000 is true by construction.

2. Intervals. Earlier: "every confidence interval is a cluster bootstrap over cases." The slope intervals resampled runs. Over scenarios, two of four exclude zero and the second aggregator model excludes zero in neither scenario set (Figure 4).

3. Head-to-head. Earlier: both judges' shares span 0.5 "with large effects excluded in both directions." The second judge's interval was $[0.083,\ 0.562]$ from 15 pairs and does not exclude a large loss.

4. Order of registration and runs. Earlier: "temporal ordering is verifiable from version history." Run records for experiments 25 to 60 were first committed on 24 September 2026, weeks after their results, and carry no timestamps.

5. Prior work. Earlier: "Every published MoA number is produced by an LLM preference judge" and "No published evaluation measures … whether a corrupted proposer corrupts the final answer." Both are false (Section 9).

6. The caveat extractor. Earlier: proposers "emit them as structured fields; a two-judge extraction retains only sentences both judges agree carry qualification." Proposers write prose, and the assembled system uses a document-level prompt that returns at most one quoted sentence per kind per proposer (Table 5).

7. The live second-source test. Earlier: only the rise in the second value was reported. The drop in the first value was not established ($p = 0.22$; Table 6).

8. Table of main results. Earlier: intervals for differences were printed beside levels, and cells were shaded by a rule applied to rows without intervals. Table 6 replaces it.

9. The "source-selector gradient" figure. Earlier: counts from an early experiment were plotted without the answers that stated neither value (15 to 23 of 27 per arm). The figure is removed.

10. Judge agreement. Earlier: "agreement on the deployed corpus is 0.910." True as raw agreement; chance agreement there is 0.878 and kappa is 0.26.

11. Counts. Earlier: "more than fifty registered experiments" and "fourteen instrument-validity findings." There are 58 registration entries for 61 numbered experiments, and the project's list of measurement findings had twenty (twenty-two after the experiments added in v0.4).

12. Readings withdrawn or restated. "Transport of 0.11 to 0.33" as a loss (Section 3.5); "halves corrupted-value adoption", "is causal" and "$0/27$ with a clean co-source" (Section 3.3); "self-assessment predicts behavior at chance" and "corroboration is ignored" (Section 3.7); preference shares of 0.818, 0.261 and 0.317 over consistently decided pairs (Section 3.2); "at no cost" (Section 6); the before-and-after figure, which drew separate experiments as one system.

B. Changes from Draft v0.3

1. Experiments in progress. The section that listed six registered experiments now reports their outcomes (Table 9).

2. Two sources for a figure. Earlier: with two sources "the aggregator states one or the other about evenly." That described 8 items, each all-or-nothing. On 61 scored questions the answer follows the first-listed contribution about three times in four (Section 3.4).

3. Whether the reader is told. Earlier: with two sources the aggregator "did not report that two values existed." That holds for `gpt-oss:20b` (both figures in 17% of replies). Two other aggregator models showed both figures in 97% and 67% of replies.

4. Reader models. Earlier: two readers, one of which wrote the report and the other short of the registered threshold; an instruction naming the attachment; a one-sentence list. Experiment 64 removes the first two limits and reduces the third (Section 5.3).

5. The assembled system. Earlier: its effect on wrong figures "has not been measured as a system." Experiment 67 measured it for the live chain, with a correction (Section 6).

6. The comparison of caveats with ordinary sentences. Earlier: "a registered experiment now does that." It ran and is not evaluable (Section 3.5).

7. Wording. The registration of experiment 63 allowed "catches a wrong figure when the case contradicts it." This draft says the answer follows the question's figure.

C. Changes from Draft v0.4

1. Cost. Earlier: "No comparison of equal cost … was run." Experiment 68 ran two, on accuracy (Section 7).

2. Accuracy. Earlier: "We have no evidence that the pipeline gives more correct answers than one model." On public questions that leave room to improve it does ($+0.070$), and a vote of four plain answers does as well.

3. Public benchmarks. Earlier: "No public benchmark is used to evaluate the pipeline." Experiment 68 uses MMLU-Pro.

4. The assembled system. Earlier: evaluated on judge preference and, for its live chain, on a planted figure. Its chain has now been evaluated on accuracy and was less accurate than the plain pipeline.

5. Experiment 68. Earlier: "Design only; not run", with four arms. It ran with five (Table 9), and the comparison with a

vote that the design called its key test is reported as descriptive because it changes two things at once.

6. Counts. Earlier: 65 registration entries for 68 experiments; 26 predictions in the experiments after the review. Now 66 entries and 29 predictions.

D. Deviations from Registrations

The project notebook records each of these at the time it happened.

- Experiments 30 and 31: the registered instrument failed its check and was replaced; the registered thresholds were then read as sign tests.
- Experiment 44: the registration did not say over which runs judge agreement was to be computed; the reading that let the result stand was used (0.70 on analysed runs, 0.58 overall; both are reported here).
- Experiment 47: one item's values were replaced after the pilot, and the pilot check was re-evaluated after the value matcher was repaired.
- Experiment 51: the registration did not say over which items the floor check applied; it was applied to the 11 retained items (0.00) and not to all 18 (0.33).
- Experiment 53: a pilot threshold was lowered from 0.5 to 0.25 after it failed; the experiment then stopped at its checkpoint.
- Experiment 59: the check was re-evaluated after a fourth repair to the value matcher.
- Experiments 60 and 61: token budgets were raised during the experiment after reasoning exhausted them.
- Timing: one registration was committed two minutes before its first run and one amendment 33 seconds after; some notebook dates precede the commits that introduced them by up to six days.
- Experiment 62 (amended 2 October 2026, before its judge labels were scored): the author could not provide the 404 blind labels. The three predictions that needed them are not evaluable; judge-only statistics and a check against public speculation labels are reported without a bar.
- Experiment 66 (amended 2 October 2026, before any judge answer was read): the check against 120 pairs labelled by the author was replaced by a check against public human labels with the same bars and three known-value checks with bars fixed in the amendment. One of the 25 known-value pairs that decided the failed part was malformed by the script that built it; it is counted as registered, and leaving it out gives 4 of 24.
- Experiment 67 (5 October 2026): scored a second time, at the author's decision, with the text-matching check repaired to recognise figures typeset with LaTeX separators. The repair had been written and committed three days before the experiment ran and was not carried into it. First scoring: check fails at $44/61$. Second: passes at $46/61$. The statistics of the three predictions are the same in both.
- Experiment 63: the registered consequence, that the paper may say the aggregator "catches" a wrong figure, is not used. Two further aggregator models ran with a 4,096-token allowance fixed from a smoke test on an easy item; it proved too small for one of them when contributions conflict.
- Experiments 62 to 67, timing: the registrations were pushed to the public repository 50 minutes after experiment 62's judge calls had begun and before any run of the other five. Two steps took about twice their estimated time; no design was changed for that.
- Experiment 68, execution entry (9 October 2026, before any call on a test question): MMLU-Pro replaced the candidate question sets the design named (MedQA, LegalBench, FinQA or TAT-QA). The setup of three plain drafts with the aggregator was added, and the primary prediction changed from the pipeline against the vote, which differs in two things, to the pipeline against those drafts; the vote comparison is reported as descriptive. P68.3 was given a margin of -0.05, a third verdict word ("not shown") was added, and the unit of analysis is the question rather than clusters of sources, since law is almost one source. A timing and parser check on the 15 validation questions of the three sections, which are not in the test set, preceded the entry.
- Experiment 68, timing: the entry was committed 9 seconds before the first screen call and pushed to the public repository on 13 October, after the screen and during the runs. The screen took 81 hours against 30 to 50 projected, and business stopped at its cap with 36 questions. The runs of setups (a) to (c) took 70 hours against 35, and the chain 60.5 hours, including a stall of about two hours while a second program used the model server; no data were affected.
- Experiment 68, after the screen and before any result: a check leaving out five questions kept on unreadable replies was fixed and is reported beside the registered results.
- Experiment 68, the chain: as registered, its final prompt adds the request for an answer line, and the screen and caveat list of the design were not run (Section 7).

E. Experimental Details

Phrase-swap (41). 18 scenarios, 7 repeats, 3 arms, 378 runs. Answers averaged 43.7 to 46.6 sentences across arms. A check refused to run an arm whose prompt contained any registry phrase other than its own.

Preference (43, 43-R). 126 pairs per comparison, both orders. Judge 1 `gpt-oss:20b`; judge 2 a 30B model selected from three candidates on parse rate and consistency only. Pairs decided consistently: 44, 46, 41 (judge 1) and 33, 29, 23 (judge 2). First-listed answer chosen in 0.80 to 0.83 (judge 1) and 0.87 to 0.89 (judge 2) of single judgments. Mean answer lengths 7,757 and 7,649 characters for the pipeline and single answers.

Second source (47, 59). 8 items, 5 repeats, 2 arms. Exact match on values after number-format normalisation. In experiment 59 the planning step was told which quantity to cover.

Caveats in and out (30, 46). 144 runs, 18 scenarios, two aggregator models, two scenario sets. Supply was varied by removing flagged sentences from proposer text.

Follow-up (44, 54, 56). 6 items, 36 runs per arm. Named runs: 7 (informative), 15 (none), 15 (uninformative); 21 of 36 in the live experiment. Fabrication follow-up: 120 replies.

Appended list (48). 17 scenarios with at least one flagged sentence, 7 stored answers each, 119 documents, lists of 1 to 53 sentences (median 11). **Reader (51).** 18 items written, 11 retained after the pilot check; 2 readers, 4 arms, 5 repeats, 440 reads.

Screen (55). Ten models, two screen prompts, six answers each; 18 pipeline contributions per model.

Assembled system (61). 54 runs; 108 pairs against stored plain-pipeline answers; lists of about 5 sentences.

Checkable figure (63). 61 questions kept from 66 (one integer input that occurs once and whose raised and lowered values give three answers further apart than twice the scoring tolerance), 14 templates, 5 layouts, 2 runs: 610 runs per model, temperature 0.6. `gpt-oss:20b` had a 16,384-token allowance and the two other models 4,096. No run hit the context limit. Three of the main model's runs returned an empty reply at first, after spending the whole allowance on reasoning, and were drawn again under the registered retry rule; all three are on one template. Its first-layout misses are arithmetic slips.

Readers (64). 6 readers, 11 items, 10 arms, 5 repeats: 550 reads per reader, temperature 0.8. Surrounding caveat sentences come from the item's own scenario where it has enough (four items for the list of ten) and are topped up from other scenarios elsewhere; the relevant and irrelevant arms share surroundings and position. One reader (`phi4:14b`) needed more than one try for a one-word answer in 137 of 550 reads and gave none in 14; the others answered at the first try.

Added judges (65). 378 pairs, both orders, 4 judges, 3,024 judgments at temperature 0. Every reply parsed and none hit a context limit. Pairs decided: 16 (`qwen2.5:7b`), 70 (`phi4:14b`), 0 and 0.

Conveyed (66). 285 caveat sentences matched to ordinary sentences by length within the same proposer text; 3 stored answers each; the three answer sentences closest by embedding shown to the judges; 3,420 judge calls, all parsed. The two judges agree with each other on "conveyed at all" at kappa 0.572. Check: 300 FRANK summary sentences with unanimous labels (150 without error, 150 with a content error), weighted to the pool; 50 verbatim answer sentences; 80 statements in original and altered-figure form; 50 statements from another scenario; 1,120 judge calls. Alone, judge 1 reached kappa 0.705 against the human labels and judge 2 0.480.

Live chain (67). 61 items, 2 arms, one run each; roles shuffled per item; plan and two proposer texts shared between arms; 710 model calls. Three calls returned an empty reply after exhausting a 4,096-token allowance and were drawn again. Every plan parsed at the first attempt. The follow-up went to the holder of the wrong figure in 23 of 61 redundancy runs, to the holder of the right figure in 16, to the third proposer in 12 and to nobody in 10.

Accuracy (68). Screen: 974 candidates, 4,958 direct draws, temperature 0.7, a 16,384-token allowance; 38 draws gave no readable letter (17 named two or more letters). In 85 of the 136 kept questions every wrong screen draw chose the same letter. Runs: three replicates of each setup; proposer and direct draws at temperature 0.7, aggregator at 0.6, 16,384 tokens; the chain's intermediate steps at 4,096. Unreadable replies: 17 of 1,632 direct draws, 3 and 5 of 408 for the two aggregator setups, 22 of 408 for the chain (2 whose plan could not be parsed). Mean seconds per run: direct draw 78, pipeline 270, chain about 530. In the chain no follow-up was sent in 115 of 408 runs.

F. The Sentence Instrument

Labels from	Kind	n	raw	chance	kappa
exp. 41	assumption	15,485	0.910	0.878	0.26
exp. 41	jurisdiction	15,485	0.833	0.717	0.41
exp. 41	hedging	15,485	0.828	0.763	0.28
exp. 46	all three	16,869	0.838	0.752	0.35
exp. 30/31	all three	5,697	0.833	0.772	0.27

Table 10: Agreement between the two sentence judges at the prevalence they are used at. "Chance" is the agreement two judges would reach answering independently at their own base rates.

The instrument was accepted on raw agreement against a bar of 0.70, which independent judges would pass at these base rates. It had been checked on a sample chosen to be rich in positives, where kappa was 0.51 to 0.66, and on twenty anchor sentences written within the project. Of the sentences either judge flags, both flag about one in four. On sentences that are positives by construction, both judges flag 0.66 ("modeled at") and 0.35 ("taken to be"). An earlier version that judged whole documents agreed at 0.62 and was dropped.

Experiment 62. A sample of 404 stored sentences was drawn, stratified so that about one in six carries each kind, and labelled by six local judges. The registered reference, blind labels from the author made twice a week apart, could not be provided, so the instrument remains uncompared with a person. On that sample, kappa between pairs of judges runs

from 0.27 to 0.71 with a median near 0.5. The deployed pair agrees at 0.54 to 0.61, higher than in Table 10 only because the kinds are several times more frequent in the stratified sample. Each deployed judge, shown the same sentence again in a different batch of ten, agrees with its own stored label at kappa 0.42 to 0.71, so part of the disagreement between judges is a label that moves with the batch. On 400 sentences from the SFU Review Corpus [26], where annotators marked speculation cues (modality such as "could" or "I think", which is a neighbouring construct and not the same one), the deployed rule marked 0.090 of cue sentences and 0.010 of cue-free sentences as hedging; single judges ranged from 0.155 to 0.470 and from 0.010 to 0.190. None of this is validity.

Other measurement findings recorded by the project and relevant here: an early judge prompt listed the same seven phrases the writing prompts dictated, found by a registry of every instructed phrase (125 entries); an instruction moved the dictated phrasing from $0/30$ to $28/30$ answers while other phrasings of the same idea moved from $1/30$ to $4/30$; empty judge replies had at one point been scored as negative labels; pairwise judging of two long answers is decided entirely by position for the 7 to 8B judges tried (experiment 65); and a check that looks for a figure's digits misses figures the model typesets, finding the stated figure in 94 of 122 replies of experiment 63 where 111 hold it.

G. Diagnostics That Are Not Optimized

The design records the following and does not tune toward them. *How often a disagreement is named or a follow-up sent*: raising it without new information reproduces the uninformative-reply arm of Section 3.6. *Counts of hedges or instructed phrases*: an instruction raises the phrase (Section 3.1), and the judges mark it down (Section 3.2). *The slope* w: it tracks answer length, and tuning it against the aggregator's own text would reward caveats the aggregator adds itself. *Judge preference on qualified text*: the judges mostly follow position and mark qualification down. The quantities worth optimizing are checked outside the pipeline's own text, such as exact-match correctness on items with known answers.